DevOps Lab Assessment - Experiment 9

Master Lab Record Book: All Dockerfiles, Manifests, Commands & Observations

COURSE / LAB:	DevOps Lab 2 (Containerization & K8s)	**ASSESSMENT:**	Question 9 (5 Marks)
TOPIC:	Master Compendium: Manifests, Commands & Logs	**ENVIRONMENT:**	Docker 29 / Minikube v1.36 / kubectl

1. Compendium Overview & Laboratory Index

This compendium serves as the comprehensive master record of all laboratory artifacts, configurations, declarative manifests, automation workflows, commands, and empirical observations developed throughout Lab 2 (Containerization & Kubernetes Orchestration). It centralizes all code and operational playbooks for auditing and archival in the student repository.

Exp #	Laboratory Topic	Key Artifacts Produced	Status / Outcome
1	K8s Deployment & Service	deployment.yaml, service.yaml, configmap.yaml	Verified 3 Pods & NodePort
2	Rolling Updates & Scaling	ReplicaSet scaling (3 to 5 to 2), rollout undo	Zero downtime confirmed
3	Comprehensive Technical Report	10-mark formal technical report & architecture	Complete system evaluation
4	Docker Containerization	Dockerfile (Multi-Stage), app.py, requirements	142 MB image, UID 10001
5	Docker Management & Compose	docker-compose.yml (Web + Redis + Vol + Net)	Multi-container state synced
6	Docker Storage Persistence	run_storage_demo.sh, named volumes, bind mounts	Data loss & persistence verified
7	Docker Networking & DNS	run_network_demo.sh, custom bridge net	DNS (127.0.0.11) resolution
8	Continuous Deployment CI/CD	.github/workflows/deploy.yml, Kustomize	Automated GitOps CD

2. Section 1: Complete Dockerfile Manifests

```
// File: src/app/Dockerfile

# --- Multi-Stage Production Dockerfile (src/app/Dockerfile) --- FROM python:3.11-slim AS builder WORKDIR /app COPY
requirements.txt . RUN pip install --no-cache-dir --user -r requirements.txt FROM python:3.11-slim AS runner WORKDIR /app
RUN groupadd -g 10001 appgroup && useradd -u 10001 -g appgroup -s /bin/false appuser COPY --from=builder /root/.local
/home/appuser/.local COPY app.py . RUN chown -R appuser:appgroup /app /home/appuser ENV
PATH=/home/appuser/.local/bin:$PATH PORT=5000 APP_VERSION=1.0.0 USER appuser EXPOSE 5000 CMD ["gunicorn", "--bind",
"0.0.0.0:5000", "--workers", "3", "app:app"]
```

3. Section 2: Complete Docker Compose Manifest

```
// File: src/compose/docker-compose.yml

services: web-app: build: { context: ../app, dockerfile: Dockerfile } image: devops-microservice:1.0.0 container_name:
web-app-container ports: ["5000:5000"] environment: [REDIS_HOST=cache-service, REDIS_PORT=6379] depends_on: {
cache-service: { condition: service_healthy } } networks: [app-tier-network] cache-service: image: redis:7.2-alpine
container_name: redis-cache-container volumes: [redis-data-volume:/data] networks: [app-tier-network] healthcheck: {
test: ["CMD", "redis-cli", "ping"], interval: 5s, timeout: 3s, retries: 5 } networks: { app-tier-network: { driver:
bridge, name: custom-devops-net } } volumes: { redis-data-volume: { name: custom-redis-storage } }
```

4. Section 3: Complete Kubernetes Declarative Manifests

```
// File: src/k8s/deployment.yaml & service.yaml

# --- Deployment Manifest (src/k8s/deployment.yaml) --- apiVersion: apps/v1 kind: Deployment metadata: { name: web-app-deployment, labels: { app: web-app } } spec: replicas: 3 strategy: { type: RollingUpdate, rollingUpdate: { maxSurge: 1, maxUnavailable: 0 } } selector: { matchLabels: { app: web-app } } template: metadata: { labels: { app: web-app, version: v1 } } spec: containers: - name: web-app image: devops-microservice:1.0.0 ports: [{ containerPort: 5000 }] resources: { requests: { cpu: 100m, memory: 128Mi }, limits: { cpu: 250m, memory: 256Mi } } readinessProbe: { httpGet: { path: /ready, port: 5000 }, initialDelaySeconds: 5, periodSeconds: 5 } livenessProbe: { httpGet: { path: /healthz, port: 5000 }, initialDelaySeconds: 10, periodSeconds: 10 } --- # --- Service Manifest (src/k8s/service.yaml) --- apiVersion: v1 kind: Service metadata: { name: web-app-service } spec: type: NodePort selector: { app: web-app } ports: [{ name: http, port: 80, targetPort: 5000, nodePort: 30080 }]
```

5. Section 4: Master DevOps CLI Command Reference

Domain	Command Syntax	Operational Purpose
Docker Build	docker build -t app:v1 -f Dockerfile .	Builds OCI image from Dockerfile context.
Docker Run	docker run -d -p 5000:5000 --name c1 app:v1	Runs background container with port forward.
Docker Volumes	docker run -v vol1:/data alpine	Mounts persistent named volume.
Docker Network	docker network create --driver bridge net1	Creates custom network with embedded DNS.
Docker Compose	docker compose up -d / down -v	Spins up or destroys entire multi-container stack.
K8s Apply	kubectl apply -f manifest.yaml	Declarative resource creation/update.
K8s Scaling	kubectl scale deployment/app --replicas=5	Horizontally scales pod instances.
K8s Rollout	kubectl rollout status / undo deployment/app	Tracks progress or reverts faulty revision.
K8s Diagnostics	kubectl describe pod <name> | kubectl logs <name>	Troubleshoots pod scheduling/runtime errors.

6. Section 5: Diagnostic & Troubleshooting Matrix

Failure State	Root Cause	Diagnostic Command	Remediation Action
CrashLoopBackOff	App crash on startup	kubectl logs <pod> --previous	Fix entrypoint command or missing env var.
ImagePullBackOff	Typo in tag / private repo	kubectl describe pod <pod>	Verify image tag and registry pull secret.
Pending Pod	Insufficient node CPU/RAM	kubectl describe pod | grep FailedScheduling	Lower resource requests or add node.
DNS Ping Fail	Default bridge lacks DNS	docker network inspect bridge	Migrate containers to user-defined bridge.

7. Comprehensive Summary & Conclusion

Master Record Sign-Off

This master record book encapsulates all requirements of DevOps Lab 2. All container configurations, Kubernetes manifests, and automated CI/CD pipelines adhere to modern cloud-native standards (immutability, least-privilege execution, zero-downtime rolling upgrades, persistent storage isolation, and automated self-healing). All code artifacts have been authored, validated, and structured for seamless deployment.